Experiment-7

Aim: Write a program Containing Steps to Handle Events in React

Description:

Event Handling in React

1.Create an Event Handler – Define a function that runs when an event occurs.

2.Attach to JSX Element – Use camelCase event attributes like onClick, onSubmit, onMouseOver, etc.

3.Mouse Events – onClick, onMouseOver, onMouseOut, onMouseDown, onMouseUp.

4.Keyboard Events – onKeyDown, onKeyUp.

5.Form Events – onSubmit, onChange.

6.State Management – Use useState to store and update input or dynamic values.

7.Prevent Default Behavior – event.preventDefault() stops page reload in forms.

Styling in React

1.Inline Styles – Applied directly in JSX using style={{ key: value }}; useful for dynamic styling.

External CSS (CSS File) – Create a .css file, import it, and apply styles using className.

2.CSS Modules – Use .module.css for scoped styling to avoid class name conflicts.

In cmd-D:\fsd\exp7>npx create-react-app myapp

D:\fsd\24485A0503\exp7>cd myapp

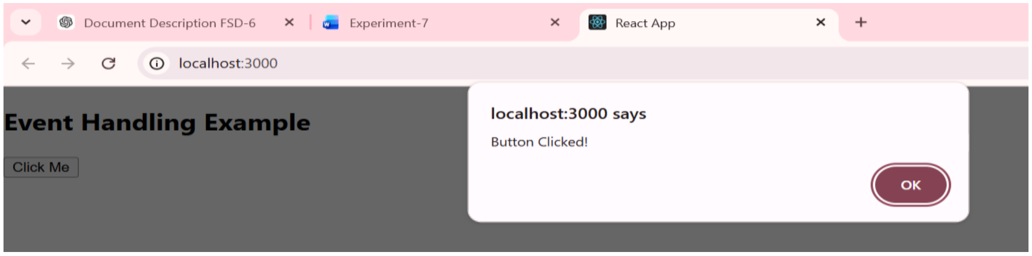
D:\fsd\24485A0503\ exp7\myapp>npm start

1: Button Click Event

Program:

```
import React from "react";
function ClickExample() {
  const handleClick = () => {
    alert("Button Clicked!");
  };
  return (
    <div>
      <h2>Event Handling Example</h2>
      <button onClick={handleClick}>Click Me</button>
    </div>
  );
}
export default ClickExample;
```

Output:

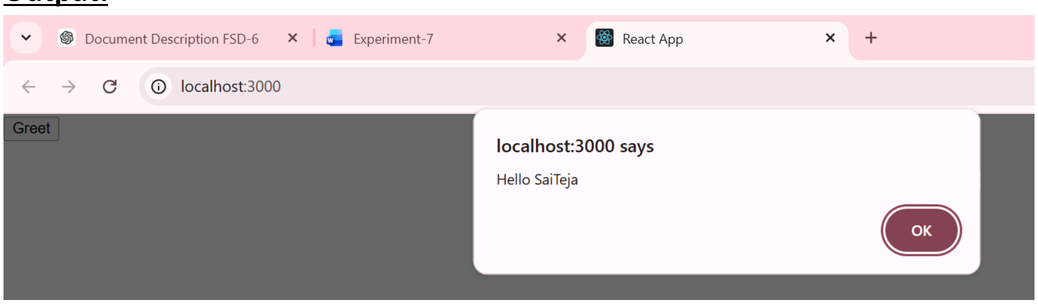


2: Passing Parameters to Event Handler

Program:

```
function App() {
  const showMessage = (name) => {
    alert("Hello " + name);
  };
  return (
    <button onClick={() => showMessage("SaiTeja")}>
      Greet
    </button>
  );
}
export default App;
```

Output:



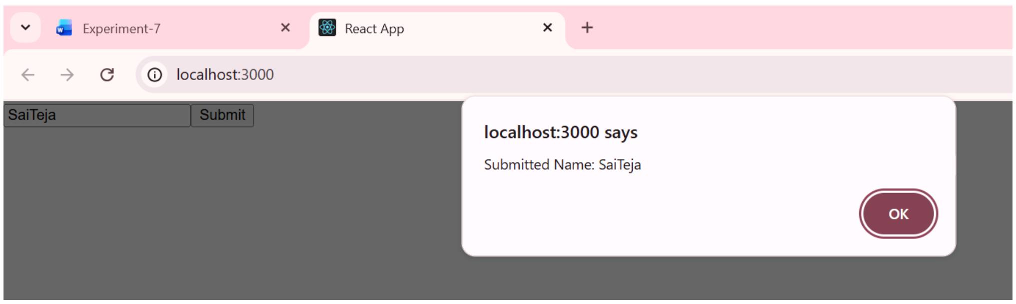
3: Handling Form Submission

Program:

```
import React, { useState } from "react";
function FormExample() {
  const [name, setName] = useState("");
  const handleSubmit = (event) => {
    event.preventDefault();   // Prevent page refresh
    alert("Submitted Name: " + name);
  };
  return (
```

```
  <form onSubmit={handleSubmit}>
    <input
      type="text"
      value={name}
      onChange={(e) => setName(e.target.value)}
      placeholder="Enter your name"
    />
    <button type="submit">Submit</button>
  </form>
 );
}
export default FormExample;
```

Output:

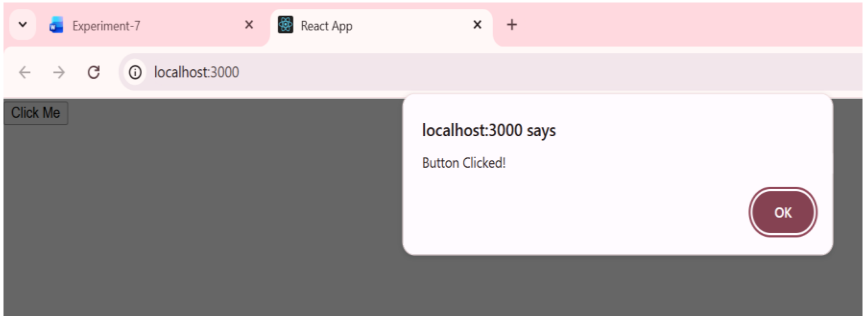


4:Mouse Event

Program:

```
import React from "react";
function MouseClickExample() {
 const handleClick = () => {
  alert("Button Clicked!");
 };
 return (
  <div>
   <button onClick={handleClick}>
    Click Me
   </button>
  </div>
 );
}
export default MouseClickExample;
```

Output:

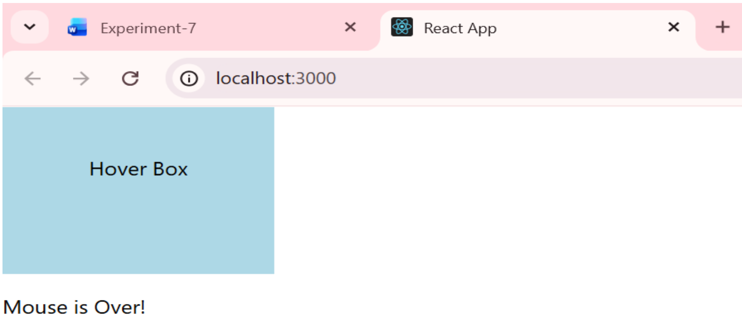


5: onMouseOver and onMouseOut (Hover Effect)

Program:

```
import React, { useState } from "react";
function HoverExample() {
  const [message, setMessage] = useState("Hover over the box");
  return (
    <div>
      <div
        style={{
          width: "200px",
          height: "100px",
          backgroundColor: "lightblue",
          textAlign: "center",
          paddingTop: "40px"
        }}
        onMouseOver={() => setMessage("Mouse is Over!")}
        onMouseOut={() => setMessage("Mouse Left!")}
      >
        Hover Box
      </div>
      <p>{message}</p>
    </div>
  );
}
export default HoverExample;
```

Output:



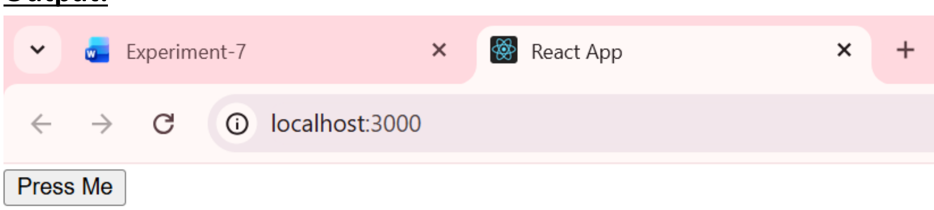
6: onMouseDown and onMouseUp

Program:

```
function MousePressExample() {
  const handleDown = () => {
    console.log("Mouse Button Pressed");
  };

  const handleUp = () => {
    console.log("Mouse Button Released");
  };
  return (
    <button onMouseDown={handleDown} onMouseUp={handleUp}>
      Press Me
    </button>
  );
}
export default MousePressExample;
```

Output:



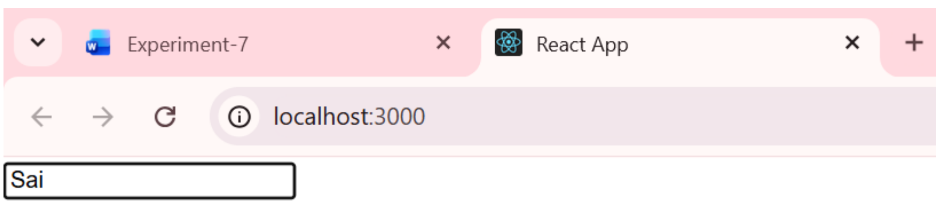
7:Key Board Events

Program:

```
import React from "react";
function KeyDownExample() {
  const handleKeyDown = (event) => {
```

```
    console.log("Key Pressed:", event.key);
  };
  return (
    <div>
      <input
        type="text"
        placeholder="Press any key"
        onKeyDown={handleKeyDown}
      />
    </div>
  );
}
 export default KeyDownExample;
```

Output:

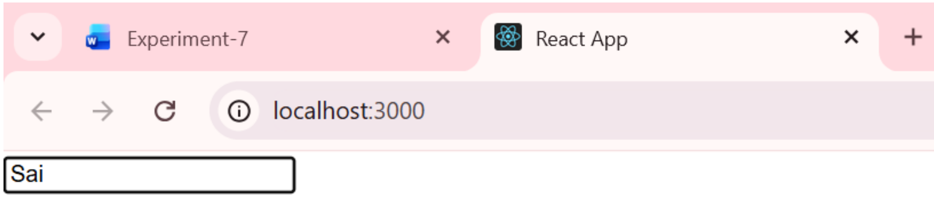


8:Key up Event

Program:

```
function KeyUpExample() {
  const handleKeyUp = (event) => {
    console.log("Key Released:", event.key);
  };
  return (
    <input
      type="text"
      placeholder="Release a key"
      onKeyUp={handleKeyUp}
    />
  );
}
export default KeyUpExample;
```

Output:

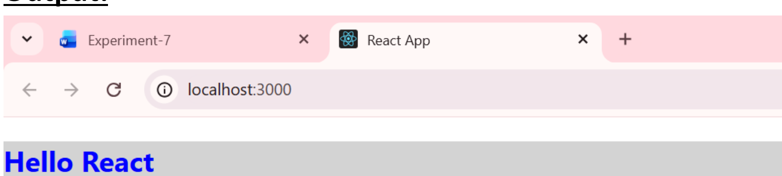


1:Styles in React js

Program:

```
import React from "react";
function InlineStyleExample() {
  return (
    <div>
      <h2 style={{ color: "blue", backgroundColor: "lightgray" }}>
        Hello React
      </h2>
    </div>
  );
}
export default InlineStyleExample;
```

Output:



2. Internal CSS (Using CSS File)

Program:

App.js

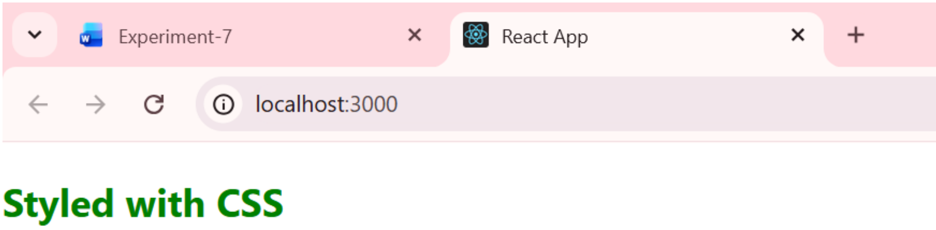
```
import React from "react";
import "./styles.css";
function CSSExample() {
  return <h2 className="title">Styled with CSS</h2>;
}
export default CSSExample;
```

App.css

```
.title {
  color: green;
```

```
  font-size: 24px;
}
```

Output:



3. CSS Modules (Scoped CSS)

Program:

App.css

```
.heading {
  color: purple;
}
```

App.js

```
import React from "react";
import styles from "./styles.module.css";

function ModuleExample() {
  return <h2 className={styles.heading}>CSS Module Example</h2>;
}
export default ModuleExample;
```

Output: